

Skills Summary

Adding and Subtracting Mixed Numbers

Use this reference while completing your worksheet or when studying.

Skill 1 — Adding mixed numbers

Plan: add the wholes, add the fraction parts, then put them back together. If the fraction parts add to more than one whole, move that extra whole over to the whole-number part.

Example: Add $1\frac{2}{5} + 2\frac{1}{5}$.

Step 1. Same-size parts (fifths), so add wholes and fifths separately, then recombine.

Step 2. Wholes: $1 + 2 = 3$; fifths: $\frac{2}{5} + \frac{1}{5} = \frac{3}{5}$; together $3\frac{3}{5}$

Try it: Add $2\frac{1}{3} + 1\frac{1}{3}$.

check: $3\frac{2}{3}$

Skill 2 — Subtracting mixed numbers

Plan: subtract the wholes, subtract the fraction parts, put them back together. Always check first whether the top fraction part is big enough — if not, you will need Skill 3.

Example: Subtract $3\frac{4}{5} - 1\frac{1}{5}$.

Step 1. $\frac{4}{5}$ is bigger than $\frac{1}{5}$, so no regrouping is needed — subtract wholes and fifths separately.

Step 2. Wholes: $3 - 1 = 2$; fifths: $\frac{4}{5} - \frac{1}{5} = \frac{3}{5}$; together $2\frac{3}{5}$

Try it: Subtract $4\frac{5}{8} - 2\frac{1}{8}$.

check: $2\frac{4}{8}$

Skill 3 — Regrouping to subtract

Rule: when the fraction you are taking away is bigger than the one you have, trade one whole for a fraction: $4\frac{1}{5} = 3\frac{6}{5}$. The value does not change — one whole is just rewritten as $\frac{5}{5}$.

Example: Subtract $4\frac{1}{5} - 1\frac{3}{5}$.

Step 1. $\frac{3}{5}$ is bigger than $\frac{1}{5}$, so trade one whole: $4\frac{1}{5}$ becomes $3\frac{6}{5}$.

Step 2. Wholes: $3 - 1 = 2$; fifths: $\frac{6}{5} - \frac{3}{5} = \frac{3}{5}$; together $2\frac{3}{5}$

Try it: Subtract $3\frac{1}{4} - 1\frac{3}{4}$.

check: $1\frac{4}{4}$

(!) Watch out: in a word problem, write the unit with your answer. An answer of $2\frac{1}{2}$ by itself does not say whether it means meters, kilograms, or cups.